# ~~Performance and limitations~~ Simultaneous measurement of ~~multidimensional OmniFold unfolding with~~ the $\nu_\mu$ charged-current inclusive cross section on hydrocarbon in five observables from MINERvA open data

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~~Applied to MINERvA open data for medium-energy~~ We report a simultaneous measurement of the $\nu_\mu$ charged-current inclusive ~~interactions, OmniFold reproduces the published muon-kinematic cross section 's integral within about 1%, with a correlated shape difference. With five simultaneous observables(adding available~~ cross section in five observables, the muon transverse and longitudinal momenta, available hadronic energy, three-momentum transfer ~~,~~ and hadronic invariant mass~~), it yields central values only, whose normalizations agree with lower-dimensional results. Pseudo-experiments measure a background-subtraction bias, a partial recovery of generator-sized truth changes, and statistical undercoverage at the tested truths. With a model-dependence envelope added linearly to the uncertainty, no tested reporting definition meets our 10% precision criterion. As an illustration, simulation-calibrated tests of hybrid null hypotheses for five generator predictions all reject at the 5%familywise level, with no evidence of gain over matched coarse projections~~, unfolded without binning with OmniFold from MINERvA medium-energy open data and reported in sixteen cells with a covariance whose median per-cell uncertainty is 7.29%. Three external generator predictions compared on all sixteen cells give $\chi^2$ of 109–272 for 16 degrees of freedom, and GiBUU on 8 cells, 58. In a fake-data study, the reported uncertainty does not cover the estimator's bias at GiBUU-like truths or at a synthetic $q_3$ tilt. The same unfolding reproduces the integral of the published muon-kinematic cross section within about 1%, with a correlated shape difference.

## I. INTRODUCTION~~AND SCOPE~~

Multidifferential neutrino cross sections test interaction models across muon and hadronic observables. Their interpretation depends on correcting detector response and assessing the bias and uncertainty of the unfolding procedure. Iterative Bayesian unfolding on a binned migration matrix [1] is widely used, but the response tensor grows rapidly when additional observables are reported. OmniFold learns event weights in an unbinned feature space through classifier-based density-ratio estimation [2, 3]. This makes simultaneous higher-dimensional unfolding possible without specifying a separate binned response tensor for every projection. Unbinned unfolding has been applied to collider data, for example eight observables in deep-inelastic scattering at H1 [4] and twenty-four $Z$+jets observables at ATLAS [5]; see Ref. [6] for a survey. In neutrino physics it has been demonstrated on a public simulated T2K near-detector sample [7]. Multidifferential neutrino measurements on data reach three dimensions [8–11].

We apply the method to MINERvA medium-energy $\nu_\mu$ data from the MINERvA open-data release [12]. ~~We first reproduce the published double-differential~~ The result is a simultaneous charged-current inclusive cross section in five observables, the transverse and longitudinal muon ~~momentum, $d^2\sigma/(dp_T\,dp_\parallel)$ [13], then add~~ momenta $p_T$ and $p_\parallel$, available hadronic energy $E_{\text{avail}}$, three-momentum transfer $q_3$, and hadronic invariant mass $W$, reported in sixteen cells with its covariance (Sec. V) and compared with five generator predictions (Sec. V A). We first reproduce the published double-differential cross section in transverse and longitudinal muon momentum, $d^2\sigma/(dp_T\,dp_\parallel)$ [13], as an end-to-end test of the unfolding, and we also give three- and ~~obtain simultaneous three-, four-, and five-observable~~ four-observable central values. Reconstructed $W$ is fixed by the reconstructed $(p_T, p_\parallel, q_3)$, so the five observables are simultaneous but not independent.

~~We evaluate this implementation as a case study of multidimensional unfolding and of inference with a biased estimator . The question is whether it~~ The validation studies assess the result through the two-dimensional reproduction, background-subtraction tests, recovery of generator-sized truth changes, a coverage test and a fake-data study of estimator bias. They examine whether the unfolding gives reliable multidimensional cross sections when the simulation mismodels the truth by generator-sized amounts~~, and what validation is needed before its outputs can be interpreted. Two properties must be distinguished.~~ . The coverage of unfolded intervals is the rate at which they contain the true cross section. ~~The calibration of a test is the rate at which itrejects a true hypothesis. Calibration can be obtained without~~

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validatedintervals by simulating the whole analysis chain at the hypothesis, so that the

The result has three limitations, each stated with it. Interval coverage has not been validated. At two held-out GENIE v3 truths, a bias test finds the estimator's bias below the reported uncertainty in every cell, but does not test interval widths. In a fake-data study at 12 truths, the reported uncertainty does not cover the estimator's bias enters the simulated null distribution.A calibration obtained in this way is valid to the extent that the simulated chain reproduces the analysis of the data.at the GiBUU-like truths or at a synthetic $q_3$ tilt (Sec. IV E). The generator $\chi^2$ depends on the flux correlations: weakening only them by 10 % lowers the $\chi^2$ of the sixteen-cell predictions by 24–48 % (Sec. V A).

*Scope.* The findings are those of one OmniFold implementation on one data set, for the observables, configurations, and deformations tested, rather than of multi-dimensional OmniFold in general. Throughout, a The three- to five-observable result is a central value and four-observable results are central values without intervals of validated coverage: the output of the estimators joint-test estimator of Sec. III, with the observable definitions, simulated detector response, background treatment, and regularization given there. The paper does not deliver a validatedfive-observable cross-section measurement. The generator tests are conditional tests of simulated sixteen-cell result of Sec. V also has total uncertainties, whose coverage is not validated. The companion note also gives simulation-calibrated tests of the generator predictions; they test hybrid null hypotheses, under which the cross section equals a prediction on every cell of a fine five-observable grid, with the MINERvA Tune v1 composition below that grid, as defined in Sec. ??. They are calibrated within the implemented simulation model, whose detector model has no calorimetric recoil-response uncertainty. Under the assumptions of Sec. ??, their decision rule controls the familywise error rate of the ten decisions at 5%, on average over the nuisance priors. The rejections and are not physical exclusions of the generatorsand provide no measurement uncertainty. They show no evidence of additional discrimination beyond matched coarse projections.

Sections IV B–VI test whether the unfolded results and their intervals can be validated. These validation studies are the paper's main results, and Table ?? lists each with its configuration and test. As a secondary application, Section II describes the data, the simulation and the observables, and Sec. III the unfolding and the reporting cells. Section IV gives the uncertainties and the validation studies, from the reproduction of the published two-dimensional measurement to the fake-data tests of the sixteen-cell result, and Sec. ?? tests generator predictionsthat the published muon-kinematic measurement already disfavours [13]V gives the result and its comparison with generator predictions. Section VI discusses what limits the estimator, and Sec. VI A summarizes the principal limitations. The companion analysis note [14] gives the full validation record, the construction of the uncertainties, and the joint tests tests with their assumptions, and is cited below by section.

## II. DATA, OBSERVABLESSIMULATION, AND DOMAINSOBSERVABLES

The analysis uses the twelve MINERvA medium-energy, forward-horn-current run periods, corresponding to $1.057 \times 10^{21}$ protons on target and an average neutrino energy near 6 GeV [13], from an earlier production of the MINERvA open-data release of pre-selected analysis ntuples (AnaTuples) [12] (see Data availability). The detector, including the downstream MINOS near detector that serves as its muon spectrometer, is described in Ref. [15]. The signal is a $\nu_\mu$ charged-current interaction, with any final state, whose true vertex lies in the tracker fiducial volume ($5980 < z < 8422$ mm, inside a hexagon of apothem 850 mm) and whose muon has $\theta_\mu < 20°$, $1.5 < p_\parallel < 60$ GeV/c, and $p_T \leq 4.5$ GeV/c. Selected simulated events outside this definition, including charged-current $\nu_\mu$ events whose muon lies outside these limits, are background. The selection requires a reconstructed vertex in the same volume and a reconstructed muon with $\theta_\mu < 20°$, matched to a MINOS track of negative charge, with a dead-time requirement, as in Ref. [13]. The flux treatment follows Ref. [16]. After selection, the sample contains 4.12 million selected data events. The simulation contains 32.8 million unweighted true signal events and 20.5 million selected simulated events. The simulation is scaled to the data exposure. The simulated events are generated with GENIE 2.12.6 [17] and the MINERvA Tune v1, which reweights GENIE with a non-resonant pion reduction, a random-phase-approximation correction, and an empirical low-recoil two-particle–two-hole (2p2h) enhancement [18]. Event selection is held fixed as the classifier inputs and unfolded observables grow from $(p_T, p_\parallel)$ to $(p_T, p_\parallel, E_{\text{avail}}, q_3, W)$. Adding features does not add selected events or relax the signal definition.

The unfolded observables are defined at truth level. Besides the muon momentum components $p_T = p_\mu \sin\theta_\mu$ and $p_\parallel = p_\mu \cos\theta_\mu$, they are the three-momentum transfer $q_3 = |\vec{q}|$ from the generator's four-vectors, the hadronic invariant mass $W = \sqrt{M^2 + 2M(E_\nu - E_\mu) - Q^2}$ with the struck-nucleon mass $M$, and the available energy

$$E_{\text{avail}} = \sum_{p}(E_p - m_p) + \sum_{\pi^\pm}(E_\pi - 135\,\text{MeV}) + \sum_{\pi^0} E_\pi + \sum_{\gamma} E_\gamma, \tag{1}$$

with a defined constant of 135 MeV, rather than the charged-pion mass, in the charged-pion term. All other final-state particles, including neutrons, contribute nothing. The sums run over the generator's final-state particles after final-state interactions, in which neutral pions appear undecayed. The 135 MeV constant is inherited from the analysis framework's definition. It defines the reported ob-

servable and is used consistently by the analysis and every prediction. ~~Replacing it by the charged-pion mass would change no true-$E_{\mathrm{avail}}$ bin by more than 1.05 %.~~ Reconstructed $E_{\mathrm{avail}}$ is 1.17 times the calorimetric recoil energy in the tracker and electromagnetic calorimeter, with muon deposits removed. Reconstructed $q_3$ and $W$ instead combine the muon kinematics with a separate calorimetric estimate $q_0$ of the energy transfer.

Reconstructed $W$ is an exact function of the reconstructed $(p_T, p_\parallel, q_3)$, so the five reconstructed inputs share information and do not provide five independent constraints. The relation holds because $W^2_{\mathrm{reco}} = (\bar{M} + q_0)^2 - q^2_{3,\mathrm{reco}}$, with $\bar{M}$ an average nucleon mass, and $q_{3,\mathrm{reco}}$ increases with $q_0$ at fixed muon kinematics. This relation is exact apart from ~~the~~ a clamp at $W^2 = 0$~~, which applies to 1.5 % of simulated events~~. Only reconstructed $W$ is fixed by the other inputs. Reconstructed $E_{\mathrm{avail}}$ comes from its own calorimetric estimator and is not a function of them. At truth level, $\theta_\mu$, and with it $p_T$ and $p_\parallel$, is measured from a fixed beam axis rather than from the true neutrino direction. Truth $W$ is therefore fixed by $(p_T, p_\parallel, q_3)$ only up to that event-by-event angle ~~, of about 1 mrad,~~ and the choice of $M$. The ~~resulting residual in truth $W$ is within $\pm 2$ MeV for 68% of events. The~~ definitions are detailed in Ref. [14], Secs. 7.1 and 8.2.

The external predictions for CH are the untuned GENIE 2.12.10 without ("GENIE CV") and with ("GENIE + MEC") the Valencia 2p2h component [19], which models meson-exchange currents (MEC). The predictions for carbon are NuWro 21.09 [20] and GiBUU 2019 [21]. All are per nucleon of their target. The GENIE and NuWro predictions are reweighted exactly, event by event, to the analysis flux. The GiBUU sample stops at $E_\nu \approx 20$ GeV. The predictions compute $W$ with the neutron mass for every event ~~. In each cell where this choice was evaluated, it changes their coarse-cell predictions by at most about one Monte Carlo standard error. Its effect cannot be evaluated for GiBUU. The predictions also~~ and measure $p_T$ about their own neutrino axis~~, which differs from the analysis truth $p_T$ by $\pm 4$ MeV for 68% of events~~. Hydrogen has no quasielastic, 2p2h, or coherent $\nu_\mu$ charged-current channel, so normalizing per nucleon does not remove the difference between the targets. In GENIE's own CH sample, the carbon-only cross section per nucleon exceeds that of CH by 8.3 % for $W < 1.1$ GeV. The NuWro and GiBUU comparisons contain a target difference of this kind, which the ~~tests~~ comparison of Sec. ~~?? do~~ V A does not correct. ~~These~~ The comparisons of the predictions with the analysis definitions are given in Ref. [14], Secs. 8.1 and 8.3.

The fine grid is the analysis binning: the published $14 \times 16$ muon-momentum bins times 7 $E_{\mathrm{avail}}$, 7 $q_3$, and 6 $W$ bins. ~~The generator tests use a~~ A coarse $3^5$ grid in $(p_T, p_\parallel, E_{\mathrm{avail}}, q_3, W)$ ~~whose edges are~~ has its edges at the fine edges nearest the simulated tertiles. A coarse cell is ~~reported~~ kept if its expected reconstructed signal and its truth weight at the data exposure are both at least 1000 events. Of the coarse cells, 109 pass; these are the joint cells. The others, which hold 0.47 % of the expected signal, stay in the total only. ~~For GiBUU, whose sample lacks high-energy neutrinos, the domain is the 72 supported cells with $p_\parallel < 6$ GeV/c.~~ The construction of the coarse grid is given in Ref. [14], Sec. 11.1.

## III. ~~ESTIMATORS AND VALIDATION DESIGN~~CROSS-SECTION EXTRACTION

Each OmniFold iteration consists of two classification steps. A reconstruction-level classifier distinguishes data from simulated events and defines a likelihood-ratio weight. The event pairing transfers that weight to truth, including unmatched events, and a truth-level classifier propagates a smooth reweighting to the next iteration. Cross sections are formed by histogramming the reweighted truth events in the reporting binning and dividing by integrated flux, target count, and bin width. Events that fail reconstruction enter as unmatched truth events, so no separate efficiency correction is applied.

Backgrounds enter the reconstruction-level step through the density the classifier learns to match. In the purity-based subtraction every data event in reconstructed bin $b$ carries the weight $w_{\mathrm{pur}}(b) = \max(0,\, 1 - B_b/D_b)$, where $D_b$ and $B_b$ are the data and predicted background counts in that bin, so the target density is $D(x)\, w_{\mathrm{pur}}(b)$. The unbinned negative-weight treatment instead adds simulated background events with negative weight to the data, so the target is $D(x) - B(x)$ pointwise; a classifier refinement [22], clipped at zero, converts the signed sample into non-negative weights before the first iteration. The two targets share bin integrals wherever $B_b < D_b$ but differ inside a bin wherever the background fraction varies within it (Ref. [14], App. C).

Each classifier is trained on the full sample whose weights it then sets, without a held-out or cross-fitted split, early stopping, or class balancing; the training details are in Ref. [14], Sec. 3.1.

~~All estimator configurations~~ Both estimators use five OmniFold iterations with gradient-boosted decision-tree classifiers [23, 24]. The iteration count and classifier capacity regularize the unfolding. The nominal settings are the scikit-learn defaults, with LightGBM matched to their capacity. The refinement capacity of the joint-test estimator was chosen in simulation closure studies. No hyperparameter search was performed on data.

Table I lists the ~~configurations~~ two estimators and the results each is used for. The two-observable central value uses ~~configuration (f): the learner and seed of configuration (a)~~ LightGBM at seed 42 with negative-weight background subtraction and ~~the refinement capacity of configuration (c)~~ a refinement of 400 trees and 31 leaves. This reproducible central value is that of its own systematic-universe ensemble, so its covariance is built with the same estimator. The ~~five-observable central value uses configuration (a), with the purity-based subtraction, whose bias is given in~~ three-

TABLE I. Estimators. Both use five OmniFold iterations, gradient-boosted trees of nominal capacity (100 trees, 8 leaves) in both OmniFold steps, and refined negative-weight background subtraction; the joint-test estimator is deterministic LightGBM (fixed seeds, reproducible training), and the two-observable estimator's refinement classifier is seeded at the estimator seed plus three. Results from the two estimators are not combined. Ref. [14], Sec. 3.2 gives the full matrix, including the variants used in development.

| Estimator | Background treatment | Used for |
|---|---|---|
| joint-test estimator | negative weights; refinement 400 trees, 31 leaves | 3D and 4D central values; sixteen-cell 5D result (Sec. V); held-out shapes (Fig. 3b); precision study |
| two-observable estimator: LightGBM, seed 42 | negative weights; refinement 400 trees, 31 leaves | 2D central value and covariance (Sec. IV B) |

and four-observable central values and the sixteen-cell five-observable result (Sec. ~~IV A. The joint tests use configuration (e) ,~~ V) use the joint-test estimator, ~~whose negative-weight treatment and higher-capacity refinement remove~~ a deterministic LightGBM with the same background treatment and refinement capacity, which removes most of the bias of ~~the purity method~~ purity-based subtraction at nominal truth in the highest-$W$ cells (Sec. IV); the note's simulation-calibrated tests use the same estimator. The variants used in development, and the results that rest on them, are described in Ref. [14], Sec. 3.2.

~~Configuration (e)~~ The joint-test estimator keeps five iterations because 10, 15, or 20 iterations did not reduce its model dependence, measured as the largest residual proxy over the truth changes used in its development. The comparison and its values are given in Ref. [14], Sec. 9.4.

### A. Reporting definitions and the precision target

To include model dependence, a dedicated study defines the envelope of a cell as the largest shift of its data result when the simulation's prior is reweighted to each of five generator-sized changes of the truth, and adds it linearly to the probabilistic half-width. The envelope is a sensitivity diagnostic that tracks the regularization bias, not an interval. Our precision target is a 10 % median half-width for the total 68% interval. With the envelope added, none of the reporting definitions of a five-observable measurement tested in the study meets this target. With the joint-test estimator, the envelope alone exceeds it for the 109 coarse joint cells and for the fallback definition, the 39 supported ($E_{\mathrm{avail}}$, $W$) cells. For a coarser, non-nested partition of all five observables into 27 cells, the same estimator's envelope alone does not exceed it, but the 1.4 % normalization contained in every probabilistic half-width lifts the median above it. The study is described in Ref. [14], Sec. 9.4.

Over these definitions, the envelope has a median of about 10–13% per cell, comparable to the generator differences that such a measurement would need to resolve.

### B. Reporting cells

A five-observable cross section with total uncertainties is reported separately from those reporting definitions: the joint-test estimator's data result on 16 cells, R00–R15. They come from the 32 boxes formed by one interior edge per axis ($p_T$ 0.7, $p_\parallel$ 4.5 and $q_3$ 2 GeV/$c$; $E_{\mathrm{avail}}$ 0.8 and $W$ 1.8 GeV): each supported box with purity below 0.1 is merged into a neighbour, 4 boxes remain unreported, and every reported cell passes the support rule. The result's covariance is given in Sec. IV A.

## IV. UNCERTAINTIES AND VALIDATION

A closure test measures how well a specified deformation of the known simulated truth is recovered, with the prior, the initial truth distribution, held fixed. Coverage, how often intervals contain their target over repeated experiments, tests the estimator and the interval construction together; the coverage tested here is pointwise, for each reported quantity separately, not the stronger simultaneous coverage of all of them. The closure, deformation, and coverage studies reweight the true signal of the simulated events used as pseudo-data, which changes their reconstructed and truth distributions together and keeps the detector response at fixed truth; the prior used to unfold them stays at Tune v1. The envelope of Sec. III A instead reweights the prior used to unfold the data. Every study uses the analysis simulation's detector response~~, except the calibration of the tests, which also draws the detector bands~~.

## V. ~~TWO-DIMENSIONAL BENCHMARK~~

### A. The reported covariance

### B. ~~Comparison with the published measurement~~

The reported covariance of the sixteen-cell result is $C_{\mathrm{total}} + C_{\mathrm{resp}}$. Unlike the envelope of Sec. III A, its model term is a covariance built from three generator shifts. $C_{\mathrm{total}}$ sums the statistical, flux, interaction, detector, lateral and normalization uncertainties of the joint-test estimator and $C_{\mathrm{model}}$. $C_{\mathrm{model}}$ is built from the change in the unfolded data when the prior is reweighted to GENIE + MEC, NuWro and GiBUU in turn; it represents unscaled generator-shape dependence, not a calibrated bound, and its coverage at truths held out of its construction has not been validated. $C_{\mathrm{resp}}$ is the

response of the result to a one-standard-deviation scale of the reconstructed recoil energy in the simulation, for protons, mesons, electromagnetic and other deposits; it does not cover neutron or muon deposits, nor species, energy-dependence or resolution effects beyond a scale, and it enters only this result, its fake-data tests and its comparison in Sec. V A. The median per-cell relative uncertainty is 7.29% (6.74% without $C_{\mathrm{resp}}$). The note gives the cells, the central values, the per-group uncertainties and the correlation matrix of the reported covariance.

### B. Two-dimensional benchmark

The two-dimensional result is an end-to-end test of the unfolding against a measurement from the same data. Its integrated cross section is $3.069 \times 10^{-38}\,\mathrm{cm}^2$/nucleon, compared with $3.039 \times 10^{-38}\,\mathrm{cm}^2$/nucleon in the published analysis, a ratio of 1.010. ~~Of the 205 reported bins, 95.1 % agree within 10 %.~~ The standardized bin differences formed with the published uncertainties have an RMS of 0.535. Against the full published covariance, the descriptive covariance-weighted distance is $\chi^2/\mathrm{ndf} = 2.27$. It is computed with the pseudo-inverse over the 205 bins, which keeps all 205 eigenmodes (rank 205, condition number $1.5 \times 10^{12}$), so that ndf is that rank. It indicates a correlated shape difference, not a normalization difference, spread over many eigenmodes of the published covariance. ~~No single $p_T$ or $p_\parallel$ band carries more than 15 % of it.~~ The source of the shape difference has not been identified. Figure 1 shows the principal one-dimensional projections.

~~The distance depends on the estimator. With the purity-based subtraction, ten LightGBM training seeds give 2.64–2.75, and the exact-split estimator gives 3.66 unseeded and 3.57 in one run with a fixed seed. At the central value's seed, the refined negative-weight subtraction lowers the LightGBM distance from 2.65 to 2.27.~~ We quote the LightGBM result as the central value because its covariance uses the same estimator. ~~This learner was chosen after both comparisons were known. The benchmark is therefore~~ The distance depends on the estimator (Ref. [14], Sec. 6.4). The learner was selected after the LightGBM and exact-split comparisons with the published result, so this benchmark is a reproduction rather than an independent validation. The ~~central value and the~~ purity-subtracted exact-split ~~cross-check differ in learner and in background treatment. Their per-bin difference has~~ result differs from the central value by a median of 1.03 % per bin (maximum 17.6 %). ~~Both estimators are described in~~ ; see Ref. [14], Sec. 3.2.

### C. Two-dimensional covariance

The covariance of our two-dimensional result is the sum of four blocks, each from LightGBM unfoldings with the central value's estimator: systematic universes (187 universes), statistical (300 bootstrap replicas), training seed (five seed settings, so that its own size is poorly determined), and run period (three period groups). The resulting relative uncertainty has a median of 6.41 % per bin and is 4.76 % on the integrated cross section, compared with 6.85 % and 4.61 % for the publication. The ~~systematic-universe block dominates it bin by bin and in the integral, and the run-period block raises the per-bin median from 5.98 % to 6.41 %. The~~ published flux uncertainty (median 4.01 %) is larger than ours (3.71 %), for a reason the released inputs do not let us determine. The covariance omits the approximation in the systematic-universe unfoldings for events that fail reconstruction, the learner dependence of Sec. IV B, and the dependence on the regularization; the dependence on the prior's generator shape is reported beside it (below). The covariance has not been independently validated. It provides no calibrated significance for generator comparisons. The block sizes, the assumptions of the block sum, and the tests of the approximation are given in Ref. [14], Secs. 6.3, 4.5, and 4.6.

The statistical block, with a median statistical uncertainty of 0.676 %, is a bootstrap of 300 replicas drawn at the central value's training seed, in which the background templates are also resampled. The coverage of this block and of the total covariance has not been tested. We make no coverage claim on independent simulated samples or simultaneously over all reported bins and projections. ~~The block supersedes two earlier statistical bands, whose coverage studies are described in .~~

Data from different run periods are not mutually consistent within the other uncertainties. One period group's unfolded $p_\parallel$ distribution, taken relative to its own simulation, differs from that of the earliest periods by a step of about 7 % near 8 GeV/c. The cause is not known, and these data cannot establish which period is correct. The run-period block, built from double ratios among the three period groups (each group's unfolded result over its own simulated truth, with no group chosen as the reference), includes this inconsistency in the combined result, 2.0 % on the integrated cross section. Because the block is built from the observed differences among the periods, its size is not validated and its own consistency check cannot fail.

We report the dependence on the prior's generator shape as a separate covariance beside the four-block covariance, $C_{\mathrm{model}} = \sum_k s_k s_k^{\mathsf{T}}$. Here $s_k$ is the per-bin relative change of the result when the prior's truth distribution is reweighted by the ratio of the $(p_T, p_\parallel)$ shape of generator $k$ (GENIE with meson-exchange currents, NuWro, or GiBUU) to that of GENIE, with its normalization kept. Its median is 3.71 % per bin. It is a measure of generator-shape model dependence, not a calibrated bound on the unfolding bias. Added to the four-block covariance, it gives a total with a median of 7.90 % per bin. ~~Because the reweighting keeps the prior's normalization, $C_{\mathrm{model}}$ is a shape component. It is 0.03 % on the integrated~~

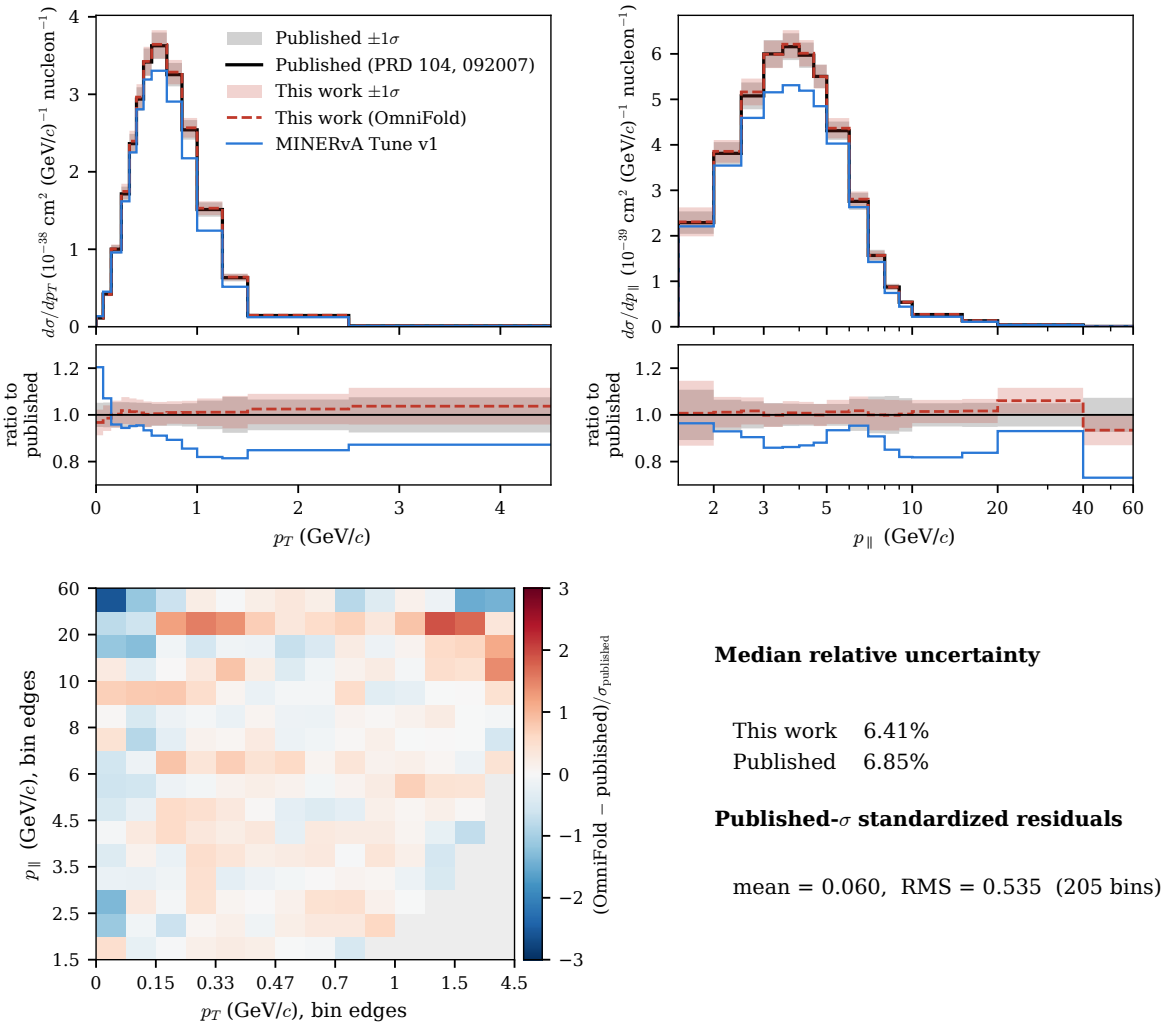



FIG. 1. Comparison with the published result [13]. Top: $p_T$ and $p_\parallel$ projections of the published result and of this analysis, each with its total $\pm1\sigma$ band (this analysis: statistical, systematic-universe (including the normalization), training-seed and run-period blocks of the covariance, Sec. IV C; it omits the learner dependence and $C_{\text{model}}$, and the two bands are correlated, so their overlap is not a compatibility test), and the MINERvA Tune v1, with ratios to the published result below. Bottom left: (OmniFold − published)/$\sigma_{\text{published}}$ per reported bin, with cells drawn at equal size (grey: not reported). Bottom right: median relative uncertainties and residual summary.

~~cross section, which leaves the integrated uncertainty at 4.76 %, and it does not test the model dependence of the normalization.~~ The comparison with the published uncertainty uses the covariance without $C_{\text{model}}$, because neither has a component for the residual dependence on the prior's generator shape: the published analysis chose its ten iterations in a fake-data study and assigns no uncertainty to the remaining dependence [13]. The iteration counts differ (ten there, five here), so we make no statement about the size of the published analysis's residual.

## V. ~~FIVE OBSERVABLES: BIAS, COVERAGE, AND PRECISION~~

### A. Normalization and closure

~~Each higher-dimensional result is checked against the lower-dimensional result it extends. In configuration (a), the four- and five-observable marginal normalizations agree at the approximately 1–2% level. For configuration (e),~~ Of the joint-test estimator's results, only the three-observable marginal has been compared with the lower-dimensional result it extends: its area-weighted total agrees with the ~~configuration-(f)~~ two-dimensional result to 0.141 %, a comparison across ~~configurations. In three observables, a configuration-(a) closure test with an injected bump in truth $E_{\text{avail}}$ leaves~~

~~a residual of 0.10 of the injected change in the $E_{\mathrm{avail}}$ marginal. In four observables , an injected bump in truth $q_3$ closes to within 1.1 % in the $q_3$ marginal, also in configuration (a). These closure tests reweight the simulated signal only.~~ the two estimators. The normalization checks of the earlier purity-subtracted five-observable central value are described in Ref. [14], Sec. 8.1, and closure checks of its estimator in three and four observables in Ref. [14], Sec. 5.1.

Figure 2 shows that, with ~~the~~ purity-based background subtraction in a deterministic LightGBM estimator, end-to-end pseudo-experiments at nominal truth reproduce the input except in the highest-$W$ cells~~. These cells~~ , which are biased by up to 4 % in simulation~~, about one quoted standard deviation in one $(E_{\mathrm{avail}}, W)$ cell. This bias is not included in the quoted uncertainties. The~~ . In the same estimator, unbinned negative-weight ~~background treatment, configuration (d),~~ subtraction with a refinement of 100 trees and 8 leaves reduces it to below 0.3 % in those cells (at most 0.4 % elsewhere in the projection), leaving up to 1 % in some joint cells. On data, the two treatments differ by at most 1.4 %. Both kinds of closure study are described in Ref. [14], Sec. 5.1.

In two observables, a closure test with backgrounds (20 pseudo-experiments at nominal truth) puts the purity and refined estimates more than one statistical standard deviation apart in 14 of the 205 cells. The purity estimate lies up to 5.9 statistical standard deviations from the truth (more than one total standard deviation in one cell), and the refined one lies within 0.44 in every cell. The two-dimensional result therefore uses the refined estimator ~~, configuration (f) (~~(Ref. [14], App. C.6).

### B. Generator-sized truth changes

Figure 3 shows that the unfolding recovers only part of a change of the $E_{\mathrm{avail}}$ shape as large as the difference between generators. When the true shape is reweighted by the GiBUU-to-GENIE ratio, the $(E_{\mathrm{avail}}, W)$ cells unfolded with ~~configuration (d)~~ a development variant of the joint-test estimator using a 100-tree, 8-leaf refinement deviate from it by several percent in the largest cells and by up to 74 % at high $W$, in the cell $0.8 \leq E_{\mathrm{avail}} < 1.5\,\mathrm{GeV}$, $W \geq 3\,\mathrm{GeV}$. These deviations occur with or without background, so background subtraction alone does not explain the residual. We interpret ~~it~~ the residual as regularization bias, because it persists over the tested ~~ranges of iteration number and classifier capacity: thirty iterations reduce the $E_{\mathrm{avail}}$ residual by 26 %, and a larger classifier improves the reconstruction-level fit without reducing it~~iteration numbers and classifier capacities, although no single cause is established. For other fixed generator shapes ~~, measured~~ tested with the joint-test estimator, it is 16 % to 31 % in the most affected $(E_{\mathrm{avail}}, W)$ cell and lies within the quoted total uncertainty in about half of those cells. ~~This total uncertainty is that of the covariance of~~ Those uncertainties come from an earlier purity-subtracted five-observable covariance that fails its specified acceptance criteria (Ref. [14], Sec. 9.6). For the sixteen-cell result, the model term of its reported covariance is the change in the unfolded data when the prior is reweighted to each generator (GENIE + MEC, NuWro, GiBUU), unscaled and not a calibrated bound; Sec. ~~IV D. No quoted uncertainty includes the regularization bias, and the five-observable covariance omits it~~IV E reports the fake-data tests of the estimator's bias. The deformation studies, the iteration and capacity tests, the exact binned comparison and the joint-test-estimator shapes are described in Ref. [14], Secs. 9.2 and 9.3.

In a binned, signal-only and noise-free diagnostic on the 109 joint cells and the truth changes used to develop the estimator, with the detector response fixed at the simulation's, an exact binned iterative unfolding with five iterations leaves about the same median residual as the boosted-tree estimator. It covers only the residuals whose diagnosis is stable under the truth resolution (38 % of the joint-cell residuals above 2 %). What unbinned information could recover, the coverage failure, and the validity of any interval lie outside its scope. Within that scope, the residual is consistent with the regularization of the inverse problem at this binning and prior, rather than with the tree approximation of the classifiers (Ref. [14], Sec. 9.3).

### C. Coverage

Statistical intervals from ~~configuration (e)~~a deterministic LightGBM estimator with purity-based subtraction, corrected for the estimator's closure bias at nominal truth, miscover at the tested truths, most strongly when the true signal rate is lowered by about 10 %. ~~The end-to-end test unfolds 400 pseudo-experiments per truth. The tested change reweights the true signal, without renormalization and with the background unchanged, and is in effect a 10 % reduction of the signal rate with a weak $E_{\mathrm{avail}}$ slope. Figure ?? shows that under that~~ Under that change, 51 of the 153 reported cell and projection integrals have a 68% coverage whose upper confidence bound lies below the acceptance threshold of 0.66, with coverage as low as 0.06. The end-to-end test unfolds 400 pseudo-experiments per truth.

A diagnosis ~~that fits a Gaussian bias and spread to each quantity's 68% and 95% coverage~~ attributes the miscoverage to a residual bias that grows with $W$ and survives the nominal-truth bias correction. The source of this residual bias has not been established. ~~The rate change itself is recovered, and, apart from a minor effect, the intervals are not too narrow.~~ Agreement in the marginal normalizations ~~thus~~ coexists with substantial residual bias under the tested deformations. The interval construction and the diagnosis are described in Ref. [14], Sec. 9.5.

~~Fractional closure residual, (unfolded − truth)/truth,~~

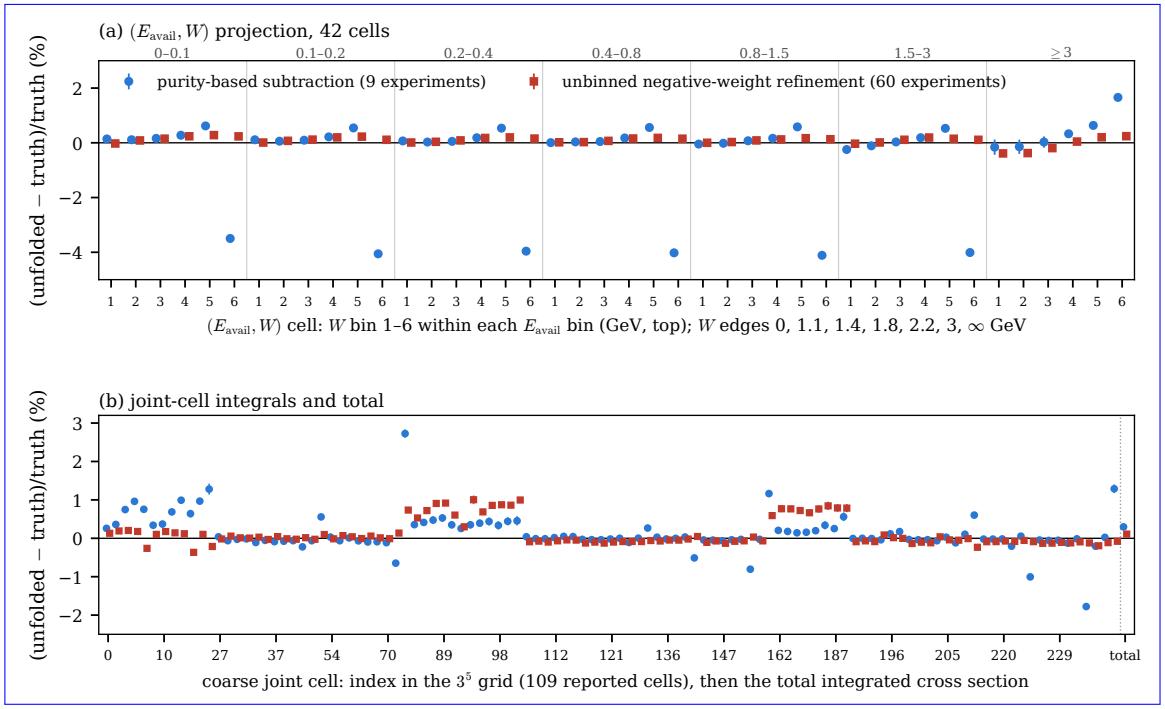



FIG. 2. Fractional closure residual, (unfolded − truth)/truth, at nominal truth, for pseudo-experiments that include backgrounds. Both from a deterministic LightGBM estimator. Circles: purity-based background subtraction (9 pseudo-experiments). Squares: unbinned negative-weight subtraction with a refinement of 100 trees and 8 leaves (60 pseudo-experiments). The pseudo-experiments are not paired. (a) The 42 $(E_{\mathrm{avail}}, W)$ projection cells, grouped by $E_{\mathrm{avail}}$ bin. (b) The 109 coarse joint cells, labelled by their index in the $3^5$ grid, then the total integrated cross section. Error bars are the standard error of the mean.

~~at nominal truth, for pseudo-experiments that include backgrounds. Circles: purity-based background subtraction, configuration (c) of Table I (9 pseudo-experiments). Squares: the unbinned negative-weight treatment, configuration (d) (60 pseudo-experiments). The pseudo-experiments are not paired. (a) The 42 $(E_{\mathrm{avail}}, W)$ projection cells, grouped by $E_{\mathrm{avail}}$ bin. (b) The 109 coarse joint cells, labelled by their index in the $3^5$ grid, then the total integrated cross section. Error bars are the standard error of the mean.~~

~~Fractional recovery residual, (unfolded − truth)/truth, per $(E_{\mathrm{avail}}, W)$ cell, where the truth is each pseudo-experiment's deformed truth; all pseudo-experiments include backgrounds. (a) Configuration (d), with the truth reweighted by the GiBUU/GENIE $E_{\mathrm{avail}}$ shape ratio (60 pseudo-experiments); the grey step is the injected change of the cell truth. At nominal truth this configuration's largest residual is 0.39 % (the squares of Fig. 2a). (b) Configuration (e), the joint-test estimator, for three shape changes not used while it was chosen (NuWro in $E_{\mathrm{avail}}$; GENIE + MEC in $(E_{\mathrm{avail}}, W)$; NuWro in $(p_T, p_\parallel, E_{\mathrm{avail}})$), 20 pseudo-experiments each; the grey band is ± the quoted total uncertainty of the $(E_{\mathrm{avail}}, W)$ projection. At nominal truth (40 pseudo-experiments) its largest residual is 0.09 %. Neither seed set was held out from the estimator's development. $W$ bin 6 is $W \geq 3$ GeV. Error bars are the standard error of the mean.~~

~~Empirical coverage of the 153 reported cell and projection integrals for statistical intervals corrected for the nominal-truth bias, from configuration (c) of Table I, with 400 pseudo-experiments, at the truth with the signal rate lowered by about 10 % and a weak $E_{\mathrm{avail}}$ slope (Sec. IV C). At nominal truth (not drawn) none lies below the 0.66 threshold, and two lie below the 0.94 threshold. Circles: 68% intervals; squares: 95% intervals. Bars are one-sided Clopper–Pearson lower and upper bounds, each at level 0.05/459, a Bonferroni correction over the 153 integrals and the three truths of the study. The third truth, a change of the $q_3$ dependence at fixed $(E_{\mathrm{avail}}, W)$, is omitted because its reweighting does not implement the intended change; the test's verdict does not depend on it. Solid lines mark the nominal coverage and dashed lines the acceptance thresholds 0.66 and 0.94; filled markers have an upper bound below the threshold. The integrals are ordered as the $(E_{\mathrm{avail}}, W)$ projection (cells 0–41 and their sum), the 109 coarse joint cells (labelled by their index in the $3^5$ grid), and the total.~~

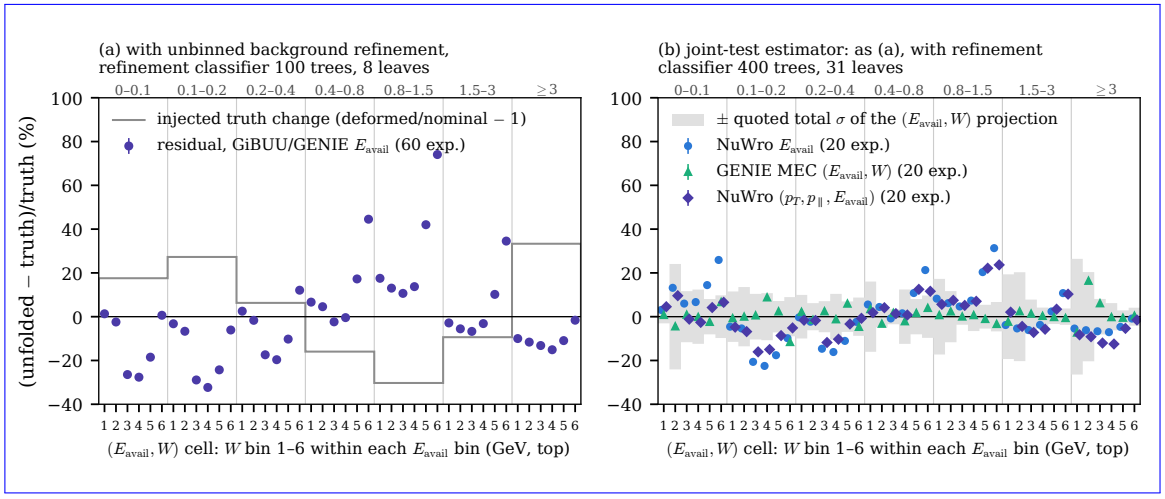



FIG. 3. Fractional recovery residual, (unfolded − truth)/truth, per $(E_{\mathrm{avail}}, W)$ cell, where the truth is each pseudo-experiment's deformed truth; all pseudo-experiments include backgrounds. (a) A development variant of the joint-test estimator, whose refinement has 100 trees and 8 leaves, with the truth reweighted by the GiBUU/GENIE $E_{\mathrm{avail}}$ shape ratio (60 pseudo-experiments); the grey step is the injected change of the cell truth. At nominal truth this variant's largest residual is 0.39 % (the squares of Fig. 2a). (b) The joint-test estimator, for three shape changes not used while it was chosen (NuWro in $E_{\mathrm{avail}}$; GENIE + MEC in $(E_{\mathrm{avail}}, W)$; NuWro in $(p_T, p_\parallel, E_{\mathrm{avail}})$), 20 pseudo-experiments each; the grey band is ± the total uncertainty of the $(E_{\mathrm{avail}}, W)$ projection of the covariance of an earlier purity-subtracted five-observable central value (Ref. [14], Sec. 9.6). At nominal truth (40 pseudo-experiments) its largest residual is 0.09 %. Neither seed set was held out from the estimator's development. $W$ bin 6 is $W \geq 3$ GeV. Error bars are the standard error of the mean.

### D. ~~Covariance and estimator~~ Estimator stability

~~The covariance of the five-observable central value, which has a 42-cell $(E_{\mathrm{avail}}, W)$ projection, does not meet the validation criteria of its specification. The comparisons of Sec. IV B use this covariance and no other five-observable covariance. The calibration of the joint tests in Sec. ?? does not use it. For one pair of seed settings, changing the estimator 's random seed moves the projected uncertainty of two further covariances by about 6.1 % in the most affected projected quantity, beyond a stability limit of 5 %; measures of the diagonal alone stay within it. We therefore quote no covariance-based generator significance. The seed dependence of the five-observable covariance rests on one seed pair, and the spread over seed pairs has not been measured. The five bands built at a fixed seed carry 26.0 % of $\sqrt{\mathrm{Tr}\, C}$, and their seed sensitivity has not been measured. The specification and the seed studies are given in .~~

~~The estimators are also~~ joint-test estimator is unstable at the level of rounding. Rounding-level ~~input changes move the results of the higher-capacity refinement~~ changes of its inputs move its results by up to 2.6 statistical standard deviations, ~~as they move those of the baseline estimator (~~and those of its lower-capacity development variant by up to 2.5~~). The refinement therefore~~, so it fails a data-stability requirement of at most 0.3 (median) and 1.0 (maximum) statistical standard deviations. ~~For the joint-test estimator, this~~ This spread lies inside the data-bootstrap statistical uncertainty, but it limits how exactly the central values can be reproduced.

### E. ~~Reporting definitions and~~ Fake-data tests of the ~~precision target~~sixteen-cell result

To ~~include model dependence, a dedicated study defines the envelope of a cell as the largest shift of its data result when the simulation's prior is reweighted to each of five generator-sized changes of the truth, and adds it linearly to the probabilistic half-width. The envelope is a sensitivity diagnostic that tracks the regularization bias , not an interval. Our precision target is a 10 % median half-width for the total 68% interval. With the envelope added, none of the reporting definitions of a five-observable measurement tested in the study meets this target. With the joint-test estimator, the envelope alone exceeds it for the 109 coarse joint cells and for the fallback definition, the 39 supported $(E_{\mathrm{avail}}, W)$ cells . For a coarser, non-nested partition of all five observables into 27 cells, the same estimator's envelope alone does not exceed it, but the 1.4 % normalization contained in every probabilistic half-width lifts the median above it. The study is described in .~~ compare the estimator's bias with the reported covariance when the truth differs from the simulation, we unfold pseudo-data at 12 truths and compare the mean unfolded result with each truth on the 16 cells through $\chi^2$ with 16 degrees of freedom. The truths are not held out: they are the simulation's prior; five hybrid truths, each a generator prediction on the fine grid with the MINERvA Tune v1 composition below

it; NuWro shape changes in $E_{\mathrm{avail}}$ and in $(p_T, p_\parallel, E_{\mathrm{avail}})$; a GENIE + MEC shape change in $(E_{\mathrm{avail}}, W)$; a GiBUU-to-GENIE $E_{\mathrm{avail}}$ ratio; a noise-free closure at the NuWro shape; and a synthetic tilt of the $q_3$ distribution at fixed $(E_{\mathrm{avail}}, W)$. For a truth from a generator that contributes to $C_{\mathrm{model}}$, the covariance omits that generator's component, as the Cov. column of Table II shows. A truth is listed when $p < 0.01$ or when a cell has $|z| > 2$. Table II gives the results under the reported covariance and under $C_{\mathrm{total}}$ alone; 6 further truths are ungraded and given in the note.

~~Over these definitions, the envelope has a median of about 10–13% per cell, comparable to the generator differences that such a measurement would need to resolve. It motivates the different route to inference of Sec. ??: after a descriptive comparison of the central values, the generator predictions are tested as hypotheses.~~

Under the reported covariance, without the GiBUU component for the two GiBUU truths, the bias is listed at the GiBUU $E_{\mathrm{avail}}$ shape (cell R06, $z = 2.01$), at the GiBUU hybrid truth ($p = 8.0 \times 10^{-29}$; cells R02, R06, R07) and at the synthetic $q_3$ tilt ($p = 6.4 \times 10^{-6}$): the reported uncertainty does not cover the fake-data bias at these GiBUU-like truths or at the $q_3$ tilt. At the other 9 truths the bias passes both criteria: no cell beyond $2\sigma$ and $p \geq 0.01$.

## V. ~~WHAT LIMITS THE ESTIMATOR~~

~~Three results point to the unfolded result retaining part of its prior. The first is the residual under a generator-sized change of the truth in Sec. IV B, which we interpret as regularization bias. The second is the envelope of Sec. III A, which tracks the same bias and alone exceeds the precision target in two of the three tested reporting definitions. The third is the upward displacement of the external nulls by the estimator's pull toward its prior, which the calibration of Sec. ?? reproduces. The miscoverage of Sec. IV C is also attributed to a residual bias, but whether that bias has the same origin is not established.~~

~~A diagnostic in simulation, summarized in Table ??, bears on the first of these results . On the same 109 joint cells and~~ Two GENIE v3 tunes, G18_10a_02_11b and AR23_20i_00_000, were generated for this test and held out of the analysis. Each reweights the simulated truth using its shape ratio to GENIE CV, constructed as for the ~~fixed truth changes used to develop the estimator, an exact binned iterative unfolding with the same simulation and five iterations leaves about the same median residual as the boosted-tree estimator. Of the 144 cell-and-truth-change residuals above 2 % whose diagnosis is stable under the truth resolution, pooled over five truth changes, the tree approximation dominates only 15. Most of the 144 (121)lie in quantities that the binned response constrains weakly at the analysis exposure, with Cramér–Rao widths above 10 %, so the regularization trades a large variance for the bias . Longer~~ $C_{\mathrm{model}}$ components. The simulated detector response is not probed. For each tune, 20 Poisson pseudo-experiments and a noise-free ~~exact iteration removes much of the residual for the GiBUU-to-GENIE change, but not for every change. The long runs do not converge, so this rests on their last iterates.~~ closure are compared with the truth using the full reported covariance. Both tunes satisfy the bias criteria: the pseudo-experiment means give $\chi^2 = 11.84$ ($p = 0.76$) and 16.86 ($p = 0.39$), respectively, with $|z| \leq 0.65$ in every cell of the means and closures. In cell R11 at G18_10a_02_11b, the bias lies below the reported uncertainty only with $C_{\mathrm{model}}$ included. The tunes share a quasielastic cross-section model; G18_10a_02_11b and the GENIE + MEC component of $C_{\mathrm{model}}$ also use Valencia-family MEC models. These tests provide partial evidence and are not independent. Omitting that $C_{\mathrm{model}}$ component gives $\chi^2 = 15.15$ ($p = 0.51$) and 24.60 ($p = 0.077$), respectively, as a descriptive comparison. The tests assess bias, not coverage: only event counts fluctuate, with systematic parameters fixed, so interval widths cannot be tested.

~~The diagnostic is binned, signal-only and noise-free, with the detector response fixed at the simulation's. It covers only the residuals whose diagnosis is stable under the truth resolution (38 % of the joint-cell residuals above 2 %) . What unbinned information could recover, the coverage failure, and the validity of any interval lie outside its scope. Within that scope, the residual is consistent with the regularization of the inverse problem at this binning and prior, rather than with the tree approximation of the classifiers ().~~

~~The studies of Secs. IV B and IV show what must be checked before unfolded intervals are interpreted: closure with~~

## V. THE SIXTEEN-CELL CROSS SECTION

### A. Result

Table III gives the sixteen-cell cross section: for each cell R00–R15, its box in the five observables, the ~~backgrounds included; truth changes of generator size, not only nominal closure; coverage under changes of the truth, not only at nominal truth; stability under training seeds and rounding-level input changes; and run-period consistency.When intervals cannot be validated, tests can still be calibrated by simulating the analysis chain at the hypothesis, as in Sec. ??~~cell-integrated cross section $f$, and its marginal relative uncertainty under the reported covariance $C_{\mathrm{total}} + C_{\mathrm{resp}}$ of Sec. IV A. The cells are correlated, so quantitative comparisons need the full covariance, whose correlation matrix is given in Ref. [14] , Table 13; Table 11 there gives the uncertainties under $C_{\mathrm{total}}$ and $C_{\mathrm{resp}}$ separately.

TABLE II. Fake-data tests on the sixteen cells: $\chi^2$ (16 degrees of freedom), $p$ and the cells with $|z| > 2$, against $C_{\text{total}} + C_{\text{resp}}$ and against $C_{\text{total}}$, each without the truth's own generator component where the second column says so (−).

| | | $C_{\text{total}} + C_{\text{resp}}$ | | | $C_{\text{total}}$ | | |
|---|---|---|---|---|---|---|---|
| Truth | Cov. | $\chi^2$ | $p$ | cells | $\chi^2$ | $p$ | cells |
| simulation prior | full | 0.00 | 1.00 | — | 0.00 | 1.00 | — |
| Tune v1, hybrid | full | 0.01 | 1.00 | — | 0.01 | 1.00 | — |
| GENIE CV, hybrid | full | 16.14 | 0.44 | — | 17.37 | 0.36 | — |
| GENIE + MEC, hybrid | −GENIE + MEC | 14.54 | 0.56 | — | 16.18 | 0.44 | — |
| GENIE + MEC, $(E_{\text{avail}}, W)$ shape | −GENIE + MEC | 0.57 | 1.00 | — | 0.77 | 1.00 | — |
| NuWro, hybrid | −NuWro | 24.92 | 0.071 | — | 29.96 | 0.018 | — |
| NuWro, $E_{\text{avail}}$ shape | −NuWro | 5.01 | 1.00 | — | 5.69 | 0.99 | — |
| NuWro, $(p_T, p_\parallel, E_{\text{avail}})$ shape | −NuWro | 30.83 | 0.014 | — | 36.19 | 0.0027 | — |
| NuWro, closure | −NuWro | 11.60 | 0.77 | — | 13.08 | 0.67 | — |
| GiBUU, $E_{\text{avail}}$ shape | −GiBUU | 25.46 | 0.062 | R06 | 35.75 | 0.0031 | R06 |
| GiBUU, hybrid | −GiBUU | 175.13 | $8.0 \times 10^{-29}$ | R02, R06, R07 | 246.75 | $< 10^{-30}$ | R02, R06, R07 |
| synthetic $q_3$ tilt | full | 53.42 | $6.4 \times 10^{-6}$ | — | 59.83 | $5.6 \times 10^{-7}$ | — |

## VI. ~~APPLICATION: COMPARISON WITH GENERATOR PREDICTIONS~~

### A. Comparison on the sixteen reported cells

Unfolded results can be compared with any prediction at truth level without simulating each generator through the detector, which is what makes the two-dimensional result reusable~~and~~, the higher-dimensional central values usable for qualitative comparison, and the sixteen-cell result usable for the comparison that follows. Table IV compares the sixteen-cell result of Sec. V with five predictions through $\chi^2$ against the reported covariance plus each prediction's finite-sample (diagonal) covariance $V_g$, with shape-only $\chi^2$ beside it. ~~For hypothesis tests the advantage is smaller.A test at reconstruction level would avoid the regularization bias, but itwould still need the same response model. Each external prediction would have to be folded through the analysis simulation's response, which is the hybrid construction of~~ Figure 4 shows the data over each prediction, cell by cell. GiBUU's sample stops at $E_\nu = 20$ GeV, so it is compared on the 8 cells with $p_\parallel < 4.5$ GeV/$c$ only. The reported uncertainty does not cover the fake-data bias at the GiBUU-like truths (Sec. ~~??, with~~ IV E). NuWro and GiBUU are on carbon, with no correction to CH, and MINERvA Tune v1 is the simulation's prior. The GENIE + MEC, NuWro and GiBUU comparisons use covariance contributions derived from those generators; GENIE CV enters as their common reference.

Weakening only the flux correlations by 10 % lowers the $\chi^2$ of the ~~same composition below the fine grid. The tests of this section instead calibrate the unfolded statistic in simulation, so that the bias enters the null distribution rather than the decision. We have not compared their power with that of a reconstruction-level test.~~ 16-cell predictions by 24–48 %. The same change to the detector or lateral correlations lowers it by at most 18 %. With the correlations of the interaction, flux, detector and lateral groups removed, every 16-cell prediction keeps $\chi^2 \geq 45$ for 16 degrees of freedom. The flux correlations are estimated from 100 hadron-production (PPFX) flux universes [16], weighted by the neutrino–electron and inverse-muon-decay constraint. Bootstrapping these universes, with the flux variances kept, gives 16th–84th percentile bands for each 16-cell $\chi^2$, with edges within 6.2 % of the reported value and a median at most 1.0 % above it. These bands describe the scatter of the $\chi^2$ from estimating the correlations with these universes, not its sensitivity to incorrect correlations, which the weakening probes; they are not intervals for the $\chi^2$ under the true correlations.

### B. Descriptive comparison of the central values

These comparisons use ~~central values only, of configuration (c) in three and four observables and configuration (a) in five~~the three- and four-observable central values of the joint-test estimator, without uncertainties. GENIE, the MINERvA Tune v1, NuWro, and GiBUU all underpredict the unfolded central value at low $E_{\text{avail}}$, between the quasielastic and Δ peaks, the region of the established low-recoil discrepancy [18, 25]. Comparisons with pion final-state-interaction variations, GENIE's Valencia 2p2h component, and the published measurement are given in Ref. [14], Secs. 7.7 and 7.8.

~~Relative to the Tune v1, two thirds of the cell-integrated positive difference in the joint~~ The five-observable $(E_{\text{avail}}, W)$ ~~projection lies at $E_{\text{avail}} \geq 0.8$ GeV and most of that at $W \geq 1.8$ GeV, but a fifth of it sits in the single widest overflow cell. In ratio, the data exceed Tune v1 by 12 % to 30 % in those cells and by 23 % to 31 % at $W < 1.1$ GeV for every $E_{\text{avail}}$, so the relative excess also extends to low $W$.~~

~~The data also lie above all four external predictions overall (data-to-prediction ratios 1.07–1.39, integrated over the projection including its overflow cells) and at high $W$. GiBUU, the most deficient, stops at $E_\nu \approx 20$ GeV, and the NuWro and GiBUU ratios contain the target difference of Sec. II. The high-$E_{\text{avail}}$, high-$W$ region is where the regularization bias of Sec. IV is largest. The map therefore describes where the central~~

TABLE III. The sixteen-cell cross section: each cell's box ($p_T$, $p_\parallel$ and $q_3$ in GeV/$c$; $E_{\rm avail}$ and $W$ in GeV), the cell-integrated cross section $f$ (an integral over the box, not a differential density) in units of $10^{-40}\,\mathrm{cm}^2$/nucleon, and its marginal relative uncertainty $\sigma/f$ (%), the square root of the diagonal of $C_{\rm total} + C_{\rm resp}$ over $f$. The largest edge on each axis is the outer edge of the analysis grid, which contains every signal event (Ref. [14], Sec. 3.5). The last row gives the median over the cells. Quantitative comparisons need the full covariance (Sec. IV A).

| cell | $p_T$ | $p_\parallel$ | $E_{\rm avail}$ | $q_3$ | $W$ | $f$ | marginal $\sigma/f$ |
|---|---|---|---|---|---|---|---|
| R00 | 0–0.7 | 1.5–4.5 | 0–0.8 | 0–2 | 0–100 | 32.6 | 7.10 |
| R01 | 0–0.7 | 1.5–4.5 | 0–100 | 2–100 | 1.8–100 | 36.7 | 7.10 |
| R02 | 0–0.7 | 1.5–4.5 | 0.8–100 | 0–2 | 0–1.8 | 6.31 | 14.00 |
| R03 | 0–0.7 | 1.5–4.5 | 0.8–100 | 0–2 | 1.8–100 | 4.30 | 12.64 |
| R04 | 0–0.7 | 4.5–60 | 0–0.8 | 0–2 | 0–100 | 49.0 | 4.20 |
| R05 | 0–0.7 | 4.5–60 | 0–100 | 2–100 | 1.8–100 | 9.98 | 7.47 |
| R06 | 0–0.7 | 4.5–60 | 0.8–100 | 0–2 | 0–1.8 | 5.57 | 5.89 |
| R07 | 0–0.7 | 4.5–60 | 0.8–100 | 0–2 | 1.8–100 | 3.21 | 7.98 |
| R08 | 0.7–4.5 | 1.5–4.5 | 0–0.8 | 0–100 | 0–1.8 | 13.9 | 6.25 |
| R09 | 0.7–4.5 | 1.5–4.5 | 0–100 | 0–100 | 1.8–100 | 39.6 | 6.26 |
| R10 | 0.7–4.5 | 1.5–4.5 | 0.8–100 | 0–2 | 0–1.8 | 7.83 | 8.59 |
| R11 | 0.7–4.5 | 1.5–4.5 | 0.8–100 | 2–100 | 0–1.8 | 5.61 | 13.69 |
| R12 | 0.7–4.5 | 4.5–60 | 0–0.8 | 0–2 | 0–100 | 22.0 | 6.63 |
| R13 | 0.7–4.5 | 4.5–60 | 0–100 | 2–100 | 0–1.8 | 9.04 | 11.07 |
| R14 | 0.7–4.5 | 4.5–60 | 0–100 | 2–100 | 1.8–100 | 49.6 | 5.27 |
| R15 | 0.7–4.5 | 4.5–60 | 0.8–100 | 0–2 | 0–100 | 11.3 | 7.94 |
| median | | | | | | | 7.29 |

TABLE IV. $\chi^2$ and its number of degrees of freedom of each prediction against the sixteen-cell result, under $C_{\rm total} + C_{\rm resp} + V_g$, with $V_g$ the prediction's finite-sample (diagonal) covariance; shape-only $\chi^2$ frees the normalization.

| Prediction | $\chi^2$ and ndf | shape-only |
|---|---|---|
| GENIE 2.12.10 CV | 109/16 | 109/15 |
| GENIE 2.12.10 + Valencia MEC | 272/16 | 258/15 |
| NuWro 21.09 | 188/16 | 188/15 |
| GiBUU 2019 | 58/8 | 28/7 |
| MINERvA Tune v1 | 330/16 | 311/15 |

~~value differs from the Tune v1. No closure test with a mismodelled detector response has been run for this map. The map and these comparisons~~ map of the earlier purity-subtracted central value, and its comparisons with these generators, are given in Ref. [14], Secs. 8.4 and 8.5.

### C. ~~Hypotheses and calibration~~

## VI. DISCUSSION AND CONCLUSIONS

~~For each of five predictions $G$ (the MINERvA Tune v1, GENIE CV, GENIE + MEC, NuWro 21.09, and GiBUU 2019), the null hypothesis $H_0(G)$ is that the true cross section equals $G$ on every cell of the fine grid of~~ The principal result is the sixteen-cell cross section of Sec. II~~. Below that grid, the null truth has the MINERvA Tune v1 simulation's event composition, varied in each pseudo-experiment by the drawn interaction-model universes with every fine-cell integral held fixed. The background is the Tune v1 simulation's, with its uncertainty bands. We call this hypothesis the hybrid null: it is the hypothesis the calibration simulates. Each prediction is tested in total and in shape on the supported coarse cells of Sec. II. Three properties are distinct. The hybrid null is defined exactly on the finite grid and needs no convergence. Whether it represents the physical generator prediction depends on the composition below the fine grid and on convergence of that grid (condition (i) below). Calibration fidelity is whether the calibration reproduces the analysis of the data (conditions (ii)–(iv)).~~ V, with its covariance and generator comparison.

~~Each calibration pseudo-experiment splits the simulation at random into two halves. The source half is reweighted per fine cell to $G$ and carries the flux, interaction-model, and detector nuisance draws. The interaction-model bands are renormalized within each fine truth cell on the signal, so that they move only the composition within the cell, and with it the efficiency and the migrations. The lateral detector bands enter as $\sum_b z_b \Delta_b$, with $\Delta_b$ from endpoint unfoldings of the data. Poisson pseudo-data are drawn from the source half, and the background pseudo-data are a Poisson draw from one half of a separate background sample. The joint-test estimator unfolds the pseudo-data with the other half of the simulation. A 1.4 % normalization, the data's rounding noise, and the prediction's finite-sample residual are then added. The data are unfolded once, with the full simulation. Because the $\Delta_b$ come from the observed data, the null distributions are conditional on these data-derived inputs, and this conditioning has not been validated ().~~

~~A total statistic $T_{\rm tot} = r^\top \Sigma^{-1} r$ is formed from the residual $r = F - \mu_G$ of the unfolded cell values $F$ and the prediction $\mu_G$. It tests the rate and the shape together, and we call it the total (rate-and-shape)~~

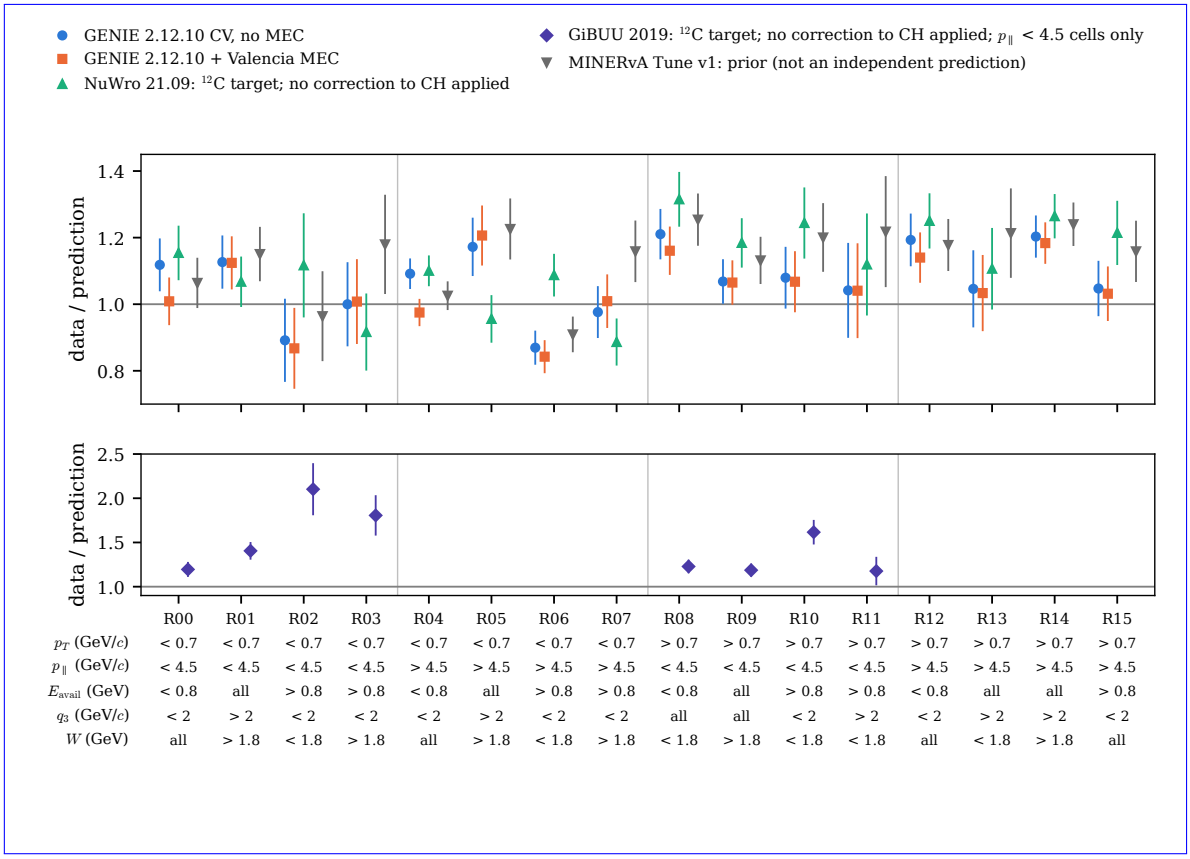



FIG. 4. Data over prediction, $f/g$, on the sixteen cells for the five predictions of Table IV. Each error bar is $(f/g)\,\sigma/f$, with $\sigma$ the square root of the diagonal of $C_{\text{total}} + C_{\text{resp}}$; the predictions' finite-sample uncertainties are not drawn. GiBUU (lower panel) is shown on its 8 cells with $p_\parallel < 4.5$ GeV/$c$ only. The key below the axes gives each cell's box. The $\chi^2$ of Table IV uses the full covariance, whose correlations the error bars do not show.

~~test. A shape statistic $T_{\text{sh}}$ of the same form uses the normalized cell fractions, with one cell dropped. The metric $\Sigma = V + \text{diag}\,\text{Var}(\mu_G)$ adds the prediction's own finite-sample variance to $V$. Here $V$ is a Ledoit–Wolf-shrunk covariance [26] of 194 pilot pseudo-experiments at the Tune v1 null, whose shrinkage floors about 80 of the 109 nearly degenerate correlation directions. The metric influences each test's power; the null calibration determines its size.~~

### A. ~~Decision rule and familywise error~~

~~The Monte Carlo $p$ value of a test is $(k+1)/(B+1)$, where $k$~~ Two validation findings point to the unfolded result retaining part of its prior. The first is the residual under a generator-sized change of ~~$B$ null pseudo-experiments reach the observed statistic; it is never zero. Each decision uses the largest $p$ value over a set of shifted nulls, which add a fixed vector to every null draw: zero, half, or all of an upper bound $D$ on the simulation-size difference along the null's bias direction (condition (iv) below), or $\pm 2\,\delta_{\text{fine}}$, where $\delta_{\text{fine}}$ is the change of the noise-free unfolded result in the last step of a refinement of the null truth's grid (condition (i)).~~

~~The ten tests are controlled at a familywise level $\alpha = 0.05$ by the Holm procedure [27], with pseudo-experiments drawn sequentially, to at most 1999 per null. A hypothesis is rejected only when the upper end of the two-sided 95% Clopper–Pearson interval [28] of its exceedance fraction lies below its Holm threshold. This rule rests on two assumptions: each true null's calibrated tail probability is uniformly distributed or conservative (A1), and the pseudo-experiments' exceedances are independent given the data (A2). Under them, the probability of meeting the criterion at any number of pseudo-experiments up to 1999 is below the threshold, whatever rule determines when sampling stops, so the decision rule controls the familywise error rate of the ten decisions at 5% (). The calibrated size is averaged over the nuisance priors: at a fixed true value of the nuisance parameters it can exceed $\alpha$. The error control applies to the decisions, not separately to the point estimates $(k+1)/(B+1)$. The calibration is valid for the simulated null to the extent that the pseudo-experiment chain reproduces the analysis of the data (A1). Every pseudo-experiment resamples the same simulated events, so measuring a rejection rate for null~~

pseudo-data processed like the data would require an independent simulation sample. No such rate has been measured.

The test design was fixed before any observed statistic was computed, after descriptive per-cell residuals of the data had been seen. The computing budget of the calibration was extended by a bounded amount after the observed $p$ values were visible. Under A1 and A2, the error control holds for any rule that decides when sampling stops, so the extension does not affect it. Of the submitted pseudo-experiments, 224 for calibration and 53 for power did not complete within the computing time limits, and the decisions use the completed ones. When rerun, none of the lost calibration pseudo-experiments reaches the observed statistic in any test or shifted null, so their outcomes change no decision. A2 also requires that the pseudo-experiments were lost independently of their outcomes, which has not been shown. Independently written code reproduces every $p$ value and decision. The design and the rerun are described in Ref. [14], Secs. and .

### A. Conditions

The decisions rest on six conditions on the nuisance model, numbered as in .

*(i) Convergence below the fine grid.* Within each fine-grid cell the null truth has the Tune v1 composition of the hypothesis definition. In the $L_2$ norm of the per-cell relative changes, the last refinement step is 0.50–0.86 of the whole change from the coarse test grid. Convergence of the fine grid has therefore not been shown, and the residual below it is unmeasured and potentially material. The shifts $\pm 2\,\delta_{\mathrm{fine}}$ were set by assuming that the refinement steps shrink geometrically. The measured $L_2$ ratios also allow slower convergence. The rejections carry over from the simulated hybrid nulls to the generator predictions themselves only if replacing the Tune v1's shape below the fine grid by $G$'s moves the noise-free unfolded result by no more than $\pm 2\,\delta_{\mathrm{fine}}$ along $\delta_{\mathrm{fine}}$. Showing this would require a demonstrated bound on the residual below the fine grid.

*(ii) Detector and hadronic response.* The detector model is the simulation's, varied only by the drawn detector bands and a linear surrogate for the lateral bands. It has no band for the calorimetric hadronic-energy response. In MINERvA's recoil-energy analyses, the uncertainty of that response is derived from test-beam calorimetry [29] and rises to 10 % at the highest three-momentum transfers of the low-recoil measurement [25]. The omission matters for the rejections. A check made after the decisions shows that the calibrated tests restricted to the matched $(p_T, p_\parallel)$ projection of the truth in Sec. IV B, which we interpret as regularization bias. The second is the envelope of Sec. ?? do not reject GENIE 2.12.10 CV ($p = 0.48$ in total and 0.63 in shape), so the information that rejects it involves the hadronic axes, on which a response band would act. The net effect of the missing band on each $p$ value has not been measured. The rejections remain conditional on the simulation's hadronic response, and their physical interpretation would require a response band drawn through the calibration.

In an exploratory sensitivity study, the simulation's reconstructed recoil energy is scaled by $\pm 4\,\%$, the level of data–simulation agreement reported for test-beam protons, pions, and electrons of 0.35–2.0 GeV/$c$ [29], and the data are re-unfolded. In this study, every rejection stays in place against the original null distributions, and the observed statistics move by up to about 11 % . The study regenerates no null distribution and tests no other form or size of response change. Neutrons and higher-momentum hadrons, which matter at high $W$, lie outside the test-beam sample. The effects the study does not test are listed in .

*(iii) Interaction-model prior.* The prior is the analysis's 34-band set, with negative weights of two bands clipped.

*(iv) Simulation size.* The pseudo-experiments use half of the simulation and the data all of it. The resulting shift enters as the upper-bound shift $D$. It is at most about 0.07 null standard deviations, measured for the mean only, without noise or nuisance draws. Repeating part of the calibration with a quarter of the simulation resolved no change of the null distributions. Under an extrapolation to the full simulation that is conservative by design, all ten decisions remain rejected. The two decisions closest to their boundary exceed the value at which they would change by 1.11 and 1.30 null standard deviations.

*(v), (vi) Domain and prediction samples.* GiBUU lacks $E_\nu > 20$ GeV, and its out-of-domain truth enters the migrations into its domain. The predictions' own finite samples enter the metric and the simulated residuals.

### A. Results

Table ?? shows that every test is rejected at familywise $\alpha = 0.05$. NuWro and GiBUU were generated on carbon. Correcting both predictions to the hydrocarbon target moves both toward the data, and NuWro's total-test rejection does not survive that correction robustly. GiBUU's deep-inelastic events on hydrogen do not conserve energy, which affects its correction and possibly its carbon prediction. The decisions of Table ?? concern each prediction on the target it was generated for . The target-corrected evaluations are descriptive and leave them unchanged. As a sensitivity check outside the decision rule, a scan of the multiplier of the shift along $\delta_{\mathrm{fine}}$ finds all ten rejections determined up to a multiplier of about 5.2. The scan probes that one direction. The calibrated null distributions differ in location: the Tune v1 null is centred at small values, whereas the external nulls are displaced far upward by the estimator's

pull toward its prior, III A, which tracks the same bias and alone exceeds the precision target in two of the three tested reporting definitions. The note's calibrated tests show the same pull: their null distributions for the external predictions are displaced upward by it, as the calibration reproduces . The target-corrected evaluations, the scan, and the null distributions with the observed statistics are given in Ref. [14], Secs. and . (Ref. [14], Sec. 11.5). The miscoverage of Sec. IV C is also attributed to a residual bias, but whether that bias has the same origin is not established.

The methodological contribution of these comparisons is a calibration that puts the estimator's bias toward its prior and the nuisance terms into each null distribution. The disagreements they recover were already established by the published two-dimensional measurement on the same data [13]. Restricted to matched $3 \times 3$ coarse projections of the same cells, onto $(p_T, p_\parallel)$ and onto $(E_{\mathrm{avail}}, W)$, the same calibrated tests give $p < 0.05$, without a multiplicity correction, for every test in at least one of the two projections. This study therefore provides no evidence of additional discrimination from the joint five-observable information. Each rejection is a global assessment of a hybrid null. Assigning a discrepancy to a kinematic region or nuclear mechanism would require separate tests. Power studies characterize the response to chosen alternatives: at the Tune v1 null, two of three specified alternatives are detected in shape with power 1.0 at the 0.005 level, and the third, a MEC-like $(E_{\mathrm{avail}}, W)$ shape, with 0.23.At the GENIE 2.12.10 null the shape powers are 0.74, 0.02, and 0.00. The projection tests and the binomial intervals of the powers are given in Ref. [14], Secs. and .

## VII. DISCUSSION AND CONCLUSIONS

The study's results are the validation findings summarized in Table ??. Section VI relates three of them to the unfolded result retaining part of its prior, and lists the checks they show to be needed The studies of Sec. IV show what must be checked before unfolded intervals are interpreted. The calibrated tests illustrate inference with a biased estimator: the decision rule controls the familywise error rate at 5% provided the calibration is exact: closure with the backgrounds included; truth changes of generator size, not only nominal closure; coverage under changes of the truth, not only at nominal truth; stability under training seeds and rounding-level input changes; and run-period consistency.

### A. Limitations

The limitations of these results are stated with the results they qualify, in Secs. IV BIII ??V, and those of the calibrated tests in the companion note. The principal ones are these. The two-dimensional covariance has not been independently validated, and its coverage has not been tested (Sec. IV C). The higher-dimensional results are central values without three- and four-observable central values lack intervals of validated coverage , and no quoted uncertainty includes their regularization bias (Sec. IV BV B). The generator tests sixteen-cell result also has total uncertainties, whose coverage is not validated; Sec. IV E reports the fake-data tests of the estimator's bias. The note's calibrated tests reject hybrid nulls, conditionally on the simulation's hadronic response. The , and the rejections carry over to the generator predictions themselves only under a bound on the residual below the fine grid, which has not been shown(Sec. ??). . Their calibration is valid to the extent that the simulated chain reproduces the analysis of the data (Sec. ??Ref. [14], Sec. 11.4).

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## DATA AVAILABILITY

The input data and simulation are the MINERvA open-data AnaTuples, released under CC0 [12]. This analysis used an earlier production of that release; its 2,374 input files are identified by SHA256 checksum in the records cited in Ref. [14], App. G. A candidate release package of about 16 MB, not yet deposited, requires only numpy, scipy, and matplotlib. It recomputes every $p$ value, decision, robustness result, and power figure of the joint tests and the numbers quoted in Sec. V B, and it regenerates

the analysis note's figures of the joint-test null distributions and the $(E_{\text{avail}}, W)$ map. The other quoted numbers, except the multiplier scan, and the event-level processing require the archived analysis products, the open-data AnaTuples, and the generator samples produced for this analysis; Ref. [14], App. G states what each part of the release does and does not reproduce. [HOLD: deposit identifier and access route, after the separately authorized release decision.]


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